

# Lecture Notes in Linear Programming - ISCD

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# Introduction

These lecture notes are exclusively destined to students of UTC. It provides fundamentals of linear programming theory and prepares students for other advanced courses in optimization and machine learning. Most of the material presented here is inspired/taken from the textbook *Linear and nonlinear Programming* [1] written by David G. Luenberger and Yinyu Ye, Stanford University, the book entitled *Linear Programming - foundations and extensions* written by Robert Vanderbei [2], Princeton University and finally *Introduction to linear optimization* [3] written by D. Bertsimas and J. Tsitsiklis, (MIT).

Our goal is to provide students with elements for a better understanding of the main ideas that lie behind the linear programming theory. We will start with basic linear programming formulations and continue with the main theoretical basis together with the solution methodology based on Simplex method. Linear programming formulations are popular because the mathematics is nicer, the theory is richer, and the computation simpler for linear problems than for nonlinear ones. In particular several modelers and solvers of high quality are available in the market.



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# Chapter 1

## Introduction to Linear Programming

### 1.1 A first example

Let start with an example: Consider a production facility (or an activity) for a manufacturing company. The facility is capable of producing a variety of products that, for simplicity, we shall enumerate as  $1, 2, \dots, n$ . These products are manufactured out of certain raw materials. Let us assume that there are  $m$  different raw materials, which again we shall simply enumerate as  $1, 2, \dots, m$ . The decisions involved in managing/operating this facility are complicated and arise dynamically as market conditions evolve around it. However, to describe a simple, fairly realistic optimization problem, we shall consider a particular snapshot of the dynamic evolution. At this specific point in time, the facility has, for each raw material  $j = 1, 2, \dots, m$ , a known amount, say  $b_j$ , on hand. Furthermore, each raw material has at this moment in time a known unit market value. We shall denote the unit value of the  $j^{th}$  raw material by  $c_j$ . In addition, each product is made from known amounts of the various raw materials. That is, producing one unit of product  $i$  requires a certain known amount, say  $a_{ij}$  units, of raw material  $j$ . Also, the  $i^{th}$  final product can be sold at the known prevailing market price of  $p_i$  euros per unit.

The problem we wish to consider is the one faced by the company's production manager. It is the problem of how to use the raw materials on hand. Let us assume that the manager decides to produce  $x_i$  units of the  $i^{th}$  product,  $i = 1, 2, \dots, n$ . As said above, the revenue associated with the production of one unit of product  $i$  is  $p_i$ . But there is also a cost of raw materials that must be considered.

The cost of producing one unit of product  $i$  is expressed as  $\sum_{j=1}^m a_{ij}c_j$ . Therefore, the net revenue associated with the production of one unit is the difference between the revenue and the cost. Since the net revenue plays an important role in our model, we introduce notation for it by setting  $f_i = p_i - \sum_{j=1}^m a_{ij}c_j$ , which gives the objectif function:

$$\max \sum_{i=1}^n f_i x_i \quad (1.1)$$

Now, let look at the raw materials on hand. Assuming that he decides to produce  $x_i$  units of the  $i^{th}$  product,  $i = 1, 2, \dots, n$ , the manager can't produce more product than he has raw material for. The amount of raw material  $j$  consumed by a given production process is  $a_{ij}$ , and so he must adhere to the following constraints:

$$\sum_{i=1}^n a_{ij} x_i \leq b_j, j = 1, 2, \dots, m \quad (1.2)$$

Last, requiring nonnegativity:

$$x_i \geq 0, i = 1, 2, \dots, n \quad (1.3)$$

To summarize, the production manager's job is to determine production values  $x_i, i = 1, 2, \dots, n$ , so as to maximize (1.1) subject to the constraints given by (1.2) and (1.3). This optimization problem is an example of a linear programming problem. This particular example is often called the resource allocation problem.

## 1.2 Linear Optimization

Optimization can be seen as the mathematical branch intended to handle a complex decision problem, involving the selection of values for a number of interrelated variables, by focusing attention on a single objective designed to quantify performance and measure the quality of the decision. The objective is maximizing (or minimizing, depending on the formulation) subject to the constraints that may limit the selection of decision variable values. All dependencies expressed through constraints or objective function are assumed linear.

In a linear program there can be distinguished:

1. Parameters, that are the data describing the instance of the problem which are considered as known given values.
2. Decision variables, that are the unknowns that the user needs to determine.
3. Objective (or optimization) function. This is used to express the objective (minimization or maximization) in terms of the decision variables.
4. Constraints are functions that express the requirements on the relationships between the decision variables.
5. Domain of variables allow to limit the set of decision variables as part of continuous/integer/positive... numbers.

**Standard form of an LP formulation.** The general mathematical programming problem can be stated as:

$$\max f(x) \quad (1.4a)$$

*subject to*

$$h_i(x) = 0, i = 1, 2, \dots, m \quad (1.4b)$$

$$g_j(x) \leq 0, j = 1, 2, \dots, p \quad (1.4c)$$

$$x \in S. \quad (1.4d)$$

Where  $x$  is an  $n$ -dimensional vector of unknowns,  $x = (x_1, x_2, \dots, x_n)$ , and  $f$ ,  $h_i, i = 1, 2, \dots, m$ , and  $g_j, j = 1, 2, \dots, p$ , are real-valued functions of the variables  $x_1, x_2, \dots, x_n$ .

The set  $S$  is a subset of  $n$ -dimensional space. The function  $f$  is the objective function of the problem and the equations, inequalities, and set restrictions are constraints. Any particular combination of decision variables is referred to as a solution. A solution is said feasible if it satisfies all constraint requirements.

In this course we will focus on a specific formulation called *standard formulation* which is specified below:

$$\max c_1x_1 + c_2x_2 + \dots + c_nx_n \quad (1.5a)$$

*subject to*

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \quad (1.5b)$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \quad (1.5c)$$

$$\dots \quad (1.5d)$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \quad (1.5e)$$

$$x_i \geq 0, \forall i \in \{1, \dots, n\}. \quad (1.5f)$$

Note that each of the constraints should be arranged such that all terms with decision variables be on the left of the constraint relation, and the only part that appears on the right is the constant term.

Converting various forms of LP to the standard form can be done as follows:

1. (Slack variables). Consider the problem as in the standard form but constraints are typically:  $a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n \leq b_i$ . These constraints can be rewritten as equalities by adding a new variable (slack variable) to the left hand side.
2. (Surplus variables). If the above linear inequalities are reversed:  $a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n \geq b_i$ . These constraints can be rewritten as equalities by subtracting a new variable (surplus variable) to the left hand side.
3. (Free variable). If  $x_i \geq 0$  is not present and hence  $x_i$  is free to take on either positive or negative values. We then write  $x_i = u_i - v_i$ , where  $u_i \geq 0$  and  $v_i \geq 0$ . An alternative for converting to standard form when  $x_i$  is unconstrained in sign is to eliminate  $x_i$  together with one of the constraint equations.

Example. Consider the following problem:

$$\max x_1 + 3x_2 + 4x_3 \quad (1.6a)$$

*subject to*

$$x_1 + 2x_2 + x_3 = 5 \quad (1.6b)$$

$$2x_1 + 3x_2 + x_3 = 6 \quad (1.6c)$$

$$x_2 \geq 0, x_3 \geq 0. \quad (1.6d)$$

Considering  $x_1 = 5 - 2x_2 - x_3$  the above can be rewritten as:

$$\max x_2 + 3x_3 \quad (1.7a)$$

*subject to*

$$x_2 + x_3 = 4 \quad (1.7b)$$

$$x_2 \geq 0, x_3 \geq 0. \quad (1.7c)$$

### Exercises

1. Convert the following problems to standard form:

a)

$$\max x + 2y + 3z \quad (1.8a)$$

*subject to*

$$2 \leq x + y \leq 3, \quad (1.8b)$$

$$6 \leq x + z \leq 15, \quad (1.8c)$$

$$x \geq 0, y \geq 0, z \geq 0 \quad (1.8d)$$

b)

$$\max x + y + z \quad (1.9a)$$

*subject to*

$$x + 2y + 3z = 10 \quad (1.9b)$$

$$x \geq 1, y \geq 2, z \geq 1. \quad (1.9c)$$

2. Convert the following problem to standard form and solve:

$$\max x_1 + 4x_2 + x_3 \quad (1.10a)$$

*subject to*

$$2x_1 - 2x_2 + x_3 = 4 \quad (1.10b)$$

$$x_1 - x_3 = 1 \quad (1.10c)$$

$$x_2 \geq 0, x_3 \geq 0. \quad (1.10d)$$

3. Convert the following problem to a linear program in standard form:

$$\min |x| + y + z \quad (1.11a)$$

*subject to*

$$x + y \leq 1 \quad (1.11b)$$

$$2x + z = 3 \quad (1.11c)$$

$$y, z \geq 0 \quad (1.11d)$$

### 1.3 Linear Program modeling

Although in the majority of cases modelling a LP problem is straightforward, in some others it may be needed to introduce new variables and/or to rearrange the constraints. We will cite a few examples below:

In some situations expressing the objective function may need using maxmin or minmax functions. Let be given variables  $x_1, x_2, \dots, x_n$ , and the objective function consists in maximizing the minimum of them:  $\max\{\min\{x_1, x_2, \dots, x_n\}\}$ . This leads to a maxmin function which cannot be expressed straightly by linear programming. A simple way to deal with it is to introduce a new variable  $y$  which is a lower bound of  $x_j$ , and state the problem as follows:

$$\max y \quad (1.12a)$$

*subject to*

$$x_j \geq y, j = 1, \dots, n \quad (1.12b)$$

...

In some other situations some *multiplicative* functions may be found. A simple example is  $\min\{1/x\}$  which may be simply expressed as  $\max\{x\}$ . In some other situations we may have bi-linear terms  $ax$  where both  $a$  and  $x$  are decision variables. Although in the general case such a formulation cannot be linearized, it can be easily handled when one of them is binary and the other is upperly bounded. Let suppose that  $x$  is a binary variable and  $a \in \{0, \dots, M\}$ . Than let first introduce a new variable  $y = ax$ . This new variable ( $y$ ) is used instead of expression  $ax$  wherever it appears in the model. Now one needs to add new constraints to make  $y$  to behave properly, that is,

$$y \leq a \quad (1.13a)$$

$$y \leq Mx \quad (1.13b)$$

$$y \geq a + (x - 1)M \quad (1.13c)$$

where  $M$  is a large value constant. The case when both are binary variables is handled similarly.

Different situations may be met when modeling constraints as well. For instance, the so-called *capacity constraints* usually give rise to upper bound constraints; In the same line, *demand constraints* will usually give rise to lower bound constraints. Another type of frequently met constraint is the so-called *mass balance* one, which expresses the requirement of conservation of mass. An obvious example of this constraint is found in flow problems when expressing that the amount of flow entering in some transit node is equal to the amount of flow going out.

Another type of constraints are the proportion constraints where one decision variable needs to represent a fixed part of a given sum. For instance, let  $a$ ,  $b$  and  $c$  give all, and  $a$  is required to be exactly 30% of all. Then, it can be expressed as:  $a = 0.3(a + b + c)$ .

## 1.4 Examples of Linear Programming problems

### 1.4.1 The knapsack problem

The knapsack problem (KP) has a significant place in the study of integer programming models with binary variables. In the knapsack problem, one needs to pack a set of items ( $I$ ), with given values ( $v$ ) and sizes ( $w$ ) (such as weights or volumes), into a container with a maximum capacity ( $C$ ). If the total size of the items exceeds the capacity, we can't pack them all. In that case, the problem is to choose a subset of the items of maximum total value that will fit in the container. The decision variables are represented by binary variables  $x_i$  taking value 1 if item  $i$  is selected and packed into the container and 0 otherwise. The objective function is  $(\max \sum_{i \in I} v_i x_i)$  such that  $(\sum_{i \in I} w_i x_i \leq C)$ .

### 1.4.2 The diet problem

We assume that there are available at the market  $n$  different foods and that the  $j^{th}$  food sells at a price  $c_j$  per unit. In addition there are  $m$  basic nutritional ingredients and, to achieve a balanced diet, each individual must receive at least  $b_i$  units of the  $i^{th}$  nutrient per day. Finally, we assume that each unit of food  $j$  contains  $a_{ij}$  units of the  $i^{th}$  nutrient.

If we denote by  $x_j$  the number of units of food  $j$  in the diet, the problem then is to select the  $x_j$ 's to minimize the total cost:  $c_1 x_1 + c_2 x_2 + \dots + c_n x_n$  subject to the nutritional constraints:  $a_{i1} x_1 + a_{i2} x_2 + \dots + a_{in} x_n \geq b_i, i = 1, \dots, m$ , and the non-negativity constraints:  $x_j \geq 0$  for all  $j$ .

### 1.4.3 The art gallery problem

The art gallery problem is a well-studied optimization problem. It originates from a real-world problem of guarding an art gallery with the minimum number of guards who together can observe the whole gallery. In the geometric version of the problem, the layout of the art gallery is represented by a simple polygon and each guard is represented by a point in the polygon. In the graph theory version the problem is represented as the vertex cover of an undirected graph problem. A vertex cover of an undirected graph  $G = (V, E)$  is a subset of its vertices such that for every edge  $(u, v)$  of the graph, either  $u$  or  $v$  is in the vertex cover. Although the name is Vertex Cover, the set covers all edges of the given graph. Given an undirected graph, the vertex cover problem is to find minimum size vertex cover. If we denote by  $x_i$  the binary variable indicating if vertex  $i$  is in the Vertex Cover or not, the problem then is to select the  $x_i$ 's to minimize:  $x_1 + x_2 + \dots + x_n$  subject to the covering constraints:  $x_i + x_j \geq 1, \forall (i, j) \in E$  and the binary constraints:  $x_i \in \{0, 1\}$  for all vertex  $i$ .

### 1.4.4 A warehousing problem

Consider the problem of operating a warehouse, by buying and selling the stock of a certain commodity, in order to maximize profit over a certain length of time, say  $n$  months. The warehouse has a fixed capacity  $C$ , and there is a cost  $r$  per unit for holding stock for one period. The selling price,  $p_i$ , of the commodity is known to fluctuate over a number of time periods  $n$  indexed by  $i$ . In any period

$i$  the price holding for purchase is  $f_i$ . The warehouse is originally empty and is required to be empty at the end of the last period.

To formulate this problem, variables are introduced for each time period. In particular, let  $z_i$  denote the level of stock in the warehouse at the beginning of period  $i$ . Let  $x_i$  denote the amount bought during period  $i$ , and let  $y_i$  denote the amount sold during period  $i$ . If there are  $n$  periods, the problem may be written as:

$$\min \sum_{i=1}^n f_i x_i - \sum_{i=1}^n p_i y_i + \sum_{i=1}^n z_i r \quad (1.14a)$$

*subject to*

$$x_i + z_i - y_i = z_{i+1} \quad 1 \leq i \leq n \quad (1.14b)$$

$$z_i \leq C \quad 1 \leq i \leq n \quad (1.14c)$$

$$z_1 = z_{n+1} = 0 \quad (1.14d)$$

$$x, y, z \geq 0. \quad (1.14e)$$

## 1.5 Exercises

1. A class of piece-wise linear functions can be represented as  $f(x) = \max(c_1^T x + d_1, c_2^T x + d_2, \dots, c_p^T x + d_p)$ . For such a function  $f$ , consider the problem minimize  $f(x)$  subject to  $Ax = b, x \geq 0$ . Show how to convert this problem to a linear programming problem

2. A manufacturer wishes to produce to produce 1000 lb of an alloy that is, by weight, 30% metal A and 70% metal B. Five alloys are available at various prices as indicated below:

| Alloy    | 1  | 2  | 3  | 4  | 5    |
|----------|----|----|----|----|------|
| %A       | 10 | 25 | 50 | 75 | 95   |
| % B      | 90 | 75 | 50 | 25 | 5    |
| Price/lb | 5  | 4  | 3  | 2  | 1.50 |

The desired alloy will be produced by combining some of the other alloys. The manufacturer wishes to find the amounts of the various alloys needed and to determine the least expensive combination. Formulate this problem as a linear program.

3. A small firm specializes in making five types of spare automobile parts. Each part is first cast from iron in the casting shop and then sent to the finishing shop where holes are drilled, surfaces are turned, and edges are ground. The required worker-hours (per 100 units) for each of the parts of the two shops are shown below:

| Part      | 1 | 2 | 3 | 4 | 5 |
|-----------|---|---|---|---|---|
| Casting   | 2 | 1 | 3 | 3 | 1 |
| Finishing | 3 | 2 | 2 | 1 | 1 |

The profits from the parts are 30, 20, 40, 25, and 10 (per 100 units), respectively. The capacities of the casting and finishing shops over the next month are

700 and 1000 worker-hours, respectively. Formulate the problem of determining the quantities of each spare part to be made during the month so as to maximize profit.

4. A small computer manufacturing company forecasts the demand over the next  $n$  months to be  $d_i$ ,  $i = 1, 2, \dots, n$ . In any month it can produce at most  $r$  units, using regular production, at a cost of  $b$  dollars per unit. By using overtime, it can produce additional units at  $c$  dollars per unit, where  $c > b$ . The firm can store units from month to month at a cost of  $s$  dollars per unit per month. Formulate the problem of determining the production schedule that minimizes the cost.



## Chapter 2

# Theoretical basis of Linear Programming

### 2.1 Basic definitions

In this chapter we will provide main basis of Linear Programming theory including definitions and main theorems. As we will see in the following, the main result shown is on the structure of an optimal solution to a standard linear programming problem. It will be shown that it suffices to look for only a certain type of solutions, called basic solutions, which limits the search to a finite, though exponential, number of solutions.

Throughout this chapter we will consider a linear program in standard form:

$$\max c'x \quad (2.1a)$$

subject to

$$Ax = b, \quad (2.1b)$$

$$x \geq 0 \quad (2.1c)$$

*Note: we use interchangeably the terms  $c'$ ,  $c^T$  to represent the transposed vector of  $c$ .*

**Basic Solutions.** Consider the system of equalities (2.1), where  $x$  is an  $n$ -vector,  $b$  an  $m$ -vector, and  $A$  is an  $m \times n$  matrix. Suppose that from the  $n$  columns of  $A$  we select a set of  $m$  linearly independent columns (such a set exists if the rank of  $A$  is  $m$ ) and denote the  $m \times m$  matrix determined by these columns by  $B$ . By putting  $x = (x_B, 0)$  we obtain a solution to  $Ax = b$ .

Recall that a set of vectors is said to be linearly dependent if one of the vectors in the set can be defined as a linear combination of the others.

**Definition 1.** A  $m \times n$  matrix  $A$  is said of full rank  $m$  if it has  $m < n$ , and the  $m$  rows of  $A$  are linearly independent.

**Definition 2.** Given the system (2.1), let  $B$  be any non-singular  $m \times m$  sub-matrix made up of columns of  $A$ . Then, if all  $n - m$  components of  $x$  not

associated with columns of  $B$  are set equal to zero, the solution to the resulting set of equations is said to be a basic solution to (2.1) with respect to the basis  $B$ . The components of  $x$  associated with columns of  $B$  are called basic variables.

**Definition 3.** If one or more of the basic variables in a basic solution has value zero, that solution is said to be a degenerate basic solution.

**Definition 4.** A vector  $x$  satisfying constraints in (2.1) is said to be feasible for these constraints. A feasible solution to (2.1) that is also basic is said to be a basic feasible solution; if this solution is also a degenerate basic solution, it is called a degenerate basic feasible solution.

Fundamental theorem of linear programming.

**Theorem 1.** Given a linear program in standard form (2.1) where  $A$  is an  $m \times n$  matrix of rank  $m$

- i) if there is a feasible solution, there is a basic feasible solution;
- ii) if there is an optimal feasible solution, there is an optimal basic feasible solution.

Proof of (i).

Let  $x$  be a feasible solution such that  $x_1a_1 + x_2a_2 + \cdots + x_na_n = b$ . Assume that exactly first  $p$  variables  $x_i$  are strictly positive. There are two cases as to whether the set  $a_1, a_2, \dots, a_p$  is linearly independent (case 1) or not (case 2).

Case 1. We have  $p \leq m$ .

If  $p = m$ , the solution is basic.

If  $p < m$ , one can build a (degenerate) basic feasible solution by assigning the value zero to  $m - p$  variables.

Case 2.  $y_1a_1 + y_2a_2 + \cdots + y_pa_p = 0$ . Multiplying this equation by a scalar  $\epsilon$  and subtracting it from  $x_1a_1 + x_2a_2 + \cdots + x_pa_p = b$  gives  $(x_1 - \epsilon y_1)a_1 + (x_2 - \epsilon y_2)a_2 + \cdots + (x_p - \epsilon y_p)a_p = b$ . For  $\epsilon = \min\{x_i/y_i : y_i > 0\}$ , the solution is feasible and has at most  $p - 1$  positive variables. We do this until obtaining a list of independent vectors where Case 1 applies.

Proof of (ii).

Let  $x = (x_1, x_2, \dots, x_n)$  be an optimal feasible solution and, as in the proof of (i), suppose there are exactly  $p$  positive variables  $x_1, x_2, \dots, x_p$ . Again there are two cases, and Case 1, corresponding to linear independence, is exactly the same as before.

Case 2. Let take again the reasoning done above in case 2 of proof for (i). It must be shown that for any  $\epsilon$  the solution is optimal. Note first that the value of the solution  $x + \epsilon y$  is  $c^T x + \epsilon c^T y$ . For  $\epsilon$  sufficiently small in magnitude,  $x + \epsilon y$  is a feasible solution for positive or negative values of  $\epsilon$ . Thus we conclude that  $c^T y = 0$ . Indeed, if  $c^T y \neq 0$ , we may find an  $\epsilon$  of small magnitude and proper sign so as to render the value of  $c^T(x + \epsilon y)$  greater than  $c^T x$  while maintaining feasibility. This would violate the assumption of optimality of  $x$  and hence we must have  $c^T y = 0$ . Next, we build the feasible basic solution as done in case 2 and note that this solution is necessarily optimal as it has the same value of the objective function.

## 2.2 Convexity

Linear programming theory is closely related to convex optimization theory. The latter one is a sub-area of optimization that studies the problem of optimizing convex functions over convex sets. The convexity makes optimization easier than the general case since reaching a local optimum comes to find a global optimum and solving the problem.

Let first recall some basic definitions on convexity.

**Definition 5.** *A real-valued function defined on an  $n$ -dimensional interval is called convex if the line segment between any two points on the graph of the function lies above or on the graph.*

Clearly, linear functions are convex.

**Definition 6.** *A convex set is a set where, for every pair of points within the set, every point on the straight line segment that joins the pair of points is also within the set.*

For instance a square or a cube define convex sets. The envelope of these objects gives what is called a polytope.

More formally, a convex polytope may be defined as the convex hull of a finite set of points (which are always bounded), or as a bounded intersection of a finite set of half-spaces.

With respect to our problem, the set defined as  $\{x \in \mathcal{R}^n | Ax \leq b\}$ ,  $A \in \mathcal{R}^{m \times n}$ ,  $b \in \mathcal{R}^m$  is called a polyhedron and  $\{x \in \mathcal{R}^n | Ax \leq b, l \leq x \leq u\}$ ,  $A \in \mathcal{R}^{m \times n}$ ,  $b \in \mathcal{R}^m$  is called a polytope. Finally,  $\{x \in \mathcal{R}^n | Ax \leq 0\}$  is called a polyhedral cone. Clearly the feasible set  $\{x \in \mathcal{R}^n | Ax \leq b\}$  of a linear program is a polyhedron.

**Definition 7.** *A point  $x$  in a convex set  $C$  is said to be an extreme point of  $C$  if there are no two distinct points  $x_1$  and  $x_2$  in  $C$  such that  $x = \alpha x_1 + (1 - \alpha)x_2$  for some  $0 < \alpha < 1$ .*

The main link between the algebraic and geometric theories is the formal relation between basic feasible solutions of linear inequalities in standard form and extreme points of polytopes. This correspondence may be established as follows:

**Theorem 2.** *Let  $A$  be an  $m \times n$  matrix of rank  $m$  and  $b$  an  $m$ -vector. Let  $K$  be the convex polytope consisting of all  $n$ -vectors  $x$  satisfying system (2.1). A vector  $x$  is an extreme point of  $K$  if and only if  $x$  is a basic feasible solution to (2.1).*

*Proof:*

Suppose first that  $x = (x_1, x_2, \dots, x_m, 0, 0, \dots, 0)$  is a basic feasible solution to (2.1). Then  $x_1 a_1 + x_2 a_2 + \dots + x_m a_m = b$ , where  $a_1, a_2, \dots, a_m$ , the first  $m$  columns of  $A$ , are linearly independent. Suppose that  $x$  could be expressed as a convex combination of two other points in  $K$ ; say,  $x = \alpha y + (1 - \alpha)z$ ,  $0 < \alpha < 1$ ,  $y \neq z$ . Since all components of  $x, y, z$  are nonnegative and since  $0 < \alpha < 1$ , it follows immediately that the last  $n - m$  components of  $y$  and  $z$  are zero. Thus,

in particular, we have

$$y_1 a_1 + y_2 a_2 + \cdots + y_m a_m = b$$

and

$$z_1 a_1 + z_2 a_2 + \cdots + z_m a_m = b$$

Since the vectors  $a_1, a_2, \dots, a_m$  are linearly independent, it follows that  $x = y = z$  and hence  $x$  is an extreme point of  $K$ . Conversely, assume that  $x$  is an extreme point of  $K$ . Let us assume that the nonzero components of  $x$  are the first  $k$  components. Then  $x_1 a_1 + x_2 a_2 + \cdots + x_k a_k = b$ , with  $x_i > 0, i = 1, 2, \dots, k$ . To show that  $x$  is a basic feasible solution it must be shown that the vectors  $a_1, a_2, \dots, a_k$  are linearly independent. We do this by contradiction. Suppose  $a_1, a_2, \dots, a_k$  are linearly dependent. Then, there is a nontrivial linear combination that is zero:

$$y_1 a_1 + y_2 a_2 + \cdots + y_k a_k = 0$$

Define the  $n$ -vector  $y = (y_1, y_2, \dots, y_k, 0, 0, \dots, 0)$ . Since  $x_i > 0, 1 \leq i \leq k$ , it is possible to select  $\epsilon$  such that

$$x + \epsilon y > 0 \quad \wedge \quad x - \epsilon y > 0.$$

We then have  $x = 1/2(x + \epsilon y) + 1/2(x - \epsilon y)$  which expresses  $x$  as a convex combination of two distinct vectors in  $K$ . This cannot occur, since  $x$  is an extreme point of  $K$ . Thus  $a_1, a_2, \dots, a_k$  are linearly independent and  $x$  is a basic feasible solution.

With respect to system of equations (2.1) we have:

**Corollary 1.** *If the convex set  $K$  corresponding to system (2.1) is nonempty, it has at least one extreme point.*

**Corollary 2.** *If there is a finite optimal solution to a linear programming problem, there is a finite optimal solution which is an extreme point of the constraint set.*

**Corollary 3.** *The constraint set  $K$  corresponding to system (2.1) possesses at most a finite number of extreme points.*

**Corollary 4.** *If the convex polytope  $K$  corresponding to system (2.1) is bounded, then  $K$  is a convex polyhedron, that is,  $K$  consists of points that are convex combinations of a finite number of points.*

Another important result in this register is the Minkowski's theorem, which can be formulated as follows:

The two above definitions are equivalent:

(a)  $P$  is a polyhedron, i.e., there exist  $A \in \mathcal{R}^{m \times n}$  and  $b \in \mathcal{R}^m$  for some  $m$  such that  $P = \{x : Ax \leq b\}$ ;

(b)  $P$  is finitely generated, i.e. there exist (finite) vectors  $v_i$  and  $r_j$  in  $\mathcal{R}^n$  such that  $P = \text{conv}(v_1, \dots, v_s) + \text{cone}(r_1, \dots, r_t)$ .

Notice here that a ray  $r \in \mathcal{R}^n$  is a ray of polyhedron  $P$  if for each  $x \in P$  the set  $\{x + r | \lambda \geq 0\}$  is contained in  $P$ . A ray  $r$  is called extreme ray if there do not exist two linearly rays  $r_1$  and  $r_2$  of  $P$  such that  $r = \frac{1}{2}(r_1 + r_2)$ .

The above theorems and the convexity theory may be combined and used to build a methodology for achieving optimal solutions. Hence, the first theorem reduces the task of solving a linear program to that of searching over basic feasible solutions. Since for a problem having  $n$  variables and  $m$  constraints there are at most  $C_n^m$  basic solutions (corresponding to the number of ways of selecting  $m$  of  $n$  columns), there are only a finite number of possibilities. On the other hand, using the second theorem, it can be easily seen that if we have a mechanism that visits extreme points (or feasible basic solutions) and attains at some point a local optimum, then this solution is necessarily a global optimum of the problem. The solution method that we will study in the next chapter, called Simplex, follows exactly this idea: it visits only feasible basic solutions until reaching a local optima.

## 2.3 Exercises

Compute the extreme points for the constraint set defined by:

a)

$$\begin{aligned}x_1 + x_2 + x_3 &= 1 \\ x_1, x_2, x_3 &\geq 0.\end{aligned}$$

Answer: There are three extreme points which gives also the three feasible basic solutions  $(1, 0, 0)$ ,  $(0, 1, 0)$  and  $(0, 0, 1)$ .

b)

$$\begin{aligned}x_1 + x_2 + x_3 &= 1 \\ 2x_1 + 3x_2 &= 1 \\ x_1, x_2, x_3 &\geq 0.\end{aligned}$$

Answer: Note that the system of equations itself has three basic solutions,  $(2, -1, 0)$ ,  $(1/2, 0, 1/2)$ ,  $(0, 1/3, 2/3)$ , the first of which is not feasible. The two other basic solutions are feasible and coincide with the two extreme points.



## Chapter 3

# Simplex

### 3.1 Pivoting

The idea of the simplex method is to proceed from one basic feasible solution (that is, one extreme point) of the constraint set of a problem in standard form to another, in such a way as to continually increase (or decrease - it depends on the objective function) the value of the objective function until an optimal solution is reached. The precedent theoretical results assure that it is sufficient to consider only basic feasible solutions in the search for an optimal feasible solution.

Simplex is an efficient method for moving among the feasible basic solutions to the optimal solution.

Let consider the standard form and look how to modify the above formulation such that a first basic solution is identified.

$$\max(c_1x_1 + c_2x_2 + \cdots + c_nx_n) \quad (3.1a)$$

*subject to*

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \quad (3.1b)$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \quad (3.1c)$$

$\dots$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \quad (3.1d)$$

$$x_i \geq 0, \forall i \in \{1, \dots, n\}. \quad (3.1e)$$

Let suppose that we need to find a basic solution  $x_B = (x_1, x_2, \dots, x_m, 0, \dots, 0)$  to system (3.1). Applying Gauss transformation we obtain:

$$\max(c_1x_1 + c_2x_2 + \cdots + c_nx_n) \quad (3.2a)$$

*subject to*

$$x_1 + \bar{a}_{1m+1}x_{m+1} + \cdots + \bar{a}_{1n}x_n = \bar{a}_{1,0} \quad (3.2b)$$

$$x_2 + \bar{a}_{2m+1}x_{m+1} + \cdots + \bar{a}_{2n}x_n = \bar{a}_{2,0} \quad (3.2c)$$

$\dots$

$$x_m + \bar{a}_{mm+1}x_{m+1} + \cdots + \bar{a}_{mn}x_n = \bar{a}_{m,0} \quad (3.2d)$$

$$x_i \geq 0, \forall i \in \{1, \dots, n\}. \quad (3.2e)$$

The above system (3.2) is called the canonical representation. Corresponding to this canonical representation of the system, the variables  $x_1, x_2, \dots, x_m$  are called basic and the other variables are nonbasic. The corresponding basic solution is then:  $x_1 = \bar{a}_{10}, x_2 = \bar{a}_{20}, \dots, x_m = \bar{a}_{m0}, x_{m+1} = \dots = x_n = 0$ .

| $x_1$   | $x_2$ | $\dots$ | $x_m$   | $x_{m+1}$        | $\dots$ | $x_n$          |                |
|---------|-------|---------|---------|------------------|---------|----------------|----------------|
| 1       | 0     | 0       | $\dots$ | $\bar{a}_{1m+1}$ | $\dots$ | $\bar{a}_{1n}$ | $\bar{a}_{10}$ |
| 0       | 1     | 0       | 0       | $\bar{a}_{2m+1}$ | $\dots$ | $\bar{a}_{2n}$ | $\bar{a}_{20}$ |
| 0       | 0     | 1       | $\dots$ |                  |         |                |                |
| $\dots$ |       |         |         |                  |         |                |                |
| 0       | 0     | 0       | 1       | $\bar{a}_{mm+1}$ | $\dots$ | $\bar{a}_{mn}$ | $\bar{a}_{m0}$ |

Let us look now at the process of pivoting itself. Suppose in the canonical system we wish to replace the basic variable  $x_p$  for some  $1 \leq p \leq m$ , by the nonbasic variable  $x_q$  with some  $m+1 \leq q \leq n$ . This can be done if and only if  $\bar{a}_{pq}$  is nonzero; it is accomplished by dividing row  $p$  by  $\bar{a}_{pq}$  to get a unit coefficient for  $x_q$  in the  $p^{th}$  equation, and then subtracting suitable multiples of row  $p$  from each of the other rows in order to get a zero coefficient for  $x_q$  in all other equations. In this way the pivot operation can generate a new basic solution from an old one by replacing one basic variable with a nonbasic variable. The element  $\bar{a}_{pq}$  in the original system is said to be the pivot element.

Example: Consider the system in canonical form:

$$x_1 + x_4 + x_5 - x_6 = 5 \quad (3.3a)$$

$$x_2 + 2x_4 - 3x_5 + x_6 = 3 \quad (3.3b)$$

$$x_3 - x_4 + 2x_5 - x_6 = -1. \quad (3.3c)$$

Find the basic solution having basic variables  $x_4, x_5, x_6$ .

| $x_1$ | $x_2$ | $x_3$ | $x_4$    | $x_5$ | $x_6$ |    |
|-------|-------|-------|----------|-------|-------|----|
| 1     | 0     | 0     | <b>1</b> | 1     | -1    | 5  |
| 0     | 1     | 0     | 2        | -3    | 1     | 3  |
| 0     | 0     | 1     | -1       | 2     | -1    | -1 |

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$     | $x_6$ |    |
|-------|-------|-------|-------|-----------|-------|----|
| 1     | 0     | 0     | 1     | 1         | -1    | 5  |
| -2    | 1     | 0     | 0     | <b>-5</b> | 3     | -7 |
| 1     | 0     | 1     | 0     | 3         | -2    | 4  |

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$       |      |
|-------|-------|-------|-------|-------|-------------|------|
| 3/5   | 1/5   | 0     | 1     | 0     | -2/5        | 18/5 |
| 2/5   | -1/5  | 0     | 0     | 1     | -3/5        | 7/5  |
| -1/5  | 3/5   | 1     | 0     | 0     | <b>-1/5</b> | -1/5 |

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |   |
|-------|-------|-------|-------|-------|-------|---|
| 1     | -1    | -2    | 1     | 0     | 0     | 4 |
| 1     | -2    | -3    | 0     | 1     | 0     | 2 |
| 1     | -3    | -5    | 0     | 0     | 1     | 1 |

### 3.1.1 How to use pivoting

According to Theorem 1, we know that it is only necessary to consider basic feasible solutions to the system  $Ax = b, x \geq 0$ . We assume  $b$  vector taking non negative values. We know that the pivot operation can generate a new basic solution from an old one by replacing one basic variable with a nonbasic variable. However, there are two issues that need to be handled: ensuring the non-negativity of the new obtained solution and guaranteeing the profit increase. Then, two points need to be clarified: Which basic variable (or vector) will leave the basis and which nonbasic variable will enter into the basis such that the new basis is feasible and with higher profit.

Let see first which should be the vector to leave basis. Assume there is  $x$  a basic feasible solution to system (3.1):

$$x_1 a_1 + x_2 a_2 + \cdots + x_m a_m = b$$

where:

| $a_1$    | $a_2$ | $\cdots$ | $a_m$    | $a_{m+1}$        | $\cdots$ | $a_n$          |                |
|----------|-------|----------|----------|------------------|----------|----------------|----------------|
| 1        | 0     | 0        | $\cdots$ | $\bar{a}_{1m+1}$ | $\cdots$ | $\bar{a}_{1n}$ | $\bar{a}_{10}$ |
| 0        | 1     | 0        | 0        | $\bar{a}_{2m+1}$ | $\cdots$ | $\bar{a}_{2n}$ | $\bar{a}_{20}$ |
| 0        | 0     | 1        | $\cdots$ |                  |          |                |                |
| $\cdots$ |       |          |          |                  |          |                |                |
| 0        | 0     | 0        | 1        | $\bar{a}_{mm+1}$ | $\cdots$ | $\bar{a}_{mn}$ | $\bar{a}_{m0}$ |

From the representation above we deduce:  $a_q = \bar{a}_{1q}a_1 + \bar{a}_{2q}a_2 + \cdots + \bar{a}_{mq}a_m$  for some  $q > m$ .

Next, for some  $\epsilon > 0$ , we obtain:

$$(x_1 - \epsilon \bar{a}_{1q})a_1 + (x_2 - \epsilon \bar{a}_{2q})a_2 + \cdots + (x_m - \epsilon \bar{a}_{mq})a_m + \epsilon a_q = b$$

Thus, for any  $\epsilon > 0$  the above formula gives  $b$  as a linear combination of at most  $m + 1$  vectors. For  $\epsilon = 0$  we have the old basic feasible solution. When  $\epsilon$  is increased from zero, the coefficient of  $a_q$  increases, and for small enough  $\epsilon$ , we achieve a feasible but nonbasic solution. The coefficients of the other vectors will either increase or decrease linearly as  $\epsilon$  is increased. To preserve feasibility, we may set  $\epsilon$  equal to the value corresponding to the first place where one (or more) of the coefficients vanishes. Such an  $\epsilon$  may be computed as:

$$\epsilon = \min_{1 \leq i \leq m} \{x_i / \bar{a}_{iq} : \bar{a}_{iq} > 0\}.$$

Let denote with  $p$  among the index  $i$  achieving the above minimization formula. In this case we obtain a new basic feasible solution, with the vector  $a_q$  replacing the vector  $a_p$ . If none of the  $\bar{a}_{iq}$ 's are positive, then all coefficients in the representation above increase (or remain constant) as  $\epsilon$  is increased, and no new basic feasible solution is obtained. We observe, however, that in this case, where none of the  $\bar{a}_{iq}$ 's are positive, there are feasible solutions to (3.1) having arbitrarily large coefficients. This means that the set  $K$  of feasible solutions to (3.1) is unbounded.

Let determine now which will be the variable to enter in the basis. This will guarantee achieving a minimum feasible solution. Indeed, the idea of the simplex method is to select the column so that the resulting new basic feasible

solution will yield a better value to the objective function than the previous one. This then provides the final missing chain in the simplex procedure. By an elementary calculation, which is derived below, it is possible to determine which vector should enter the basis so that the objective value is improved, and by another simple calculation, derived in the previous section, it is possible to then determine which vector should leave in order to maintain feasibility.

Suppose we have a basic feasible solution  $x$ :  $(x_B, 0) = (\bar{a}_{10}, \bar{a}_{20}, \dots, \bar{a}_{m0}, 0, 0, \dots, 0)$ . The value of the objective function corresponding to any solution  $x$  is  $z = c_1x_1 + c_2x_2 + \dots + c_nx_n$ , and hence for the above basic solution  $x$ , the corresponding value is  $z_0 = c_B^T x_B$  where  $c_B^T = [c_1, c_2, \dots, c_m]$ .

Let express now the basic variables as an expression of non basic variables. we obtain:

$$x_1 = \bar{a}_{10} - \sum_{j=m+1}^n \bar{a}_{1j}x_j \quad (3.4a)$$

$$x_2 = \bar{a}_{20} - \sum_{j=m+1}^n \bar{a}_{2j}x_j \quad (3.4b)$$

...

$$x_m = \bar{a}_{m0} - \sum_{j=m+1}^n \bar{a}_{mj}x_j \quad (3.4c)$$

$$(3.4d)$$

Using (3.4) we may eliminate  $x_1, x_2, \dots, x_m$  and obtain:

$$z = c^T x = z_0 + (c_{m+1} - z_{m+1})x_{m+1} + (c_{m+2} - z_{m+2})x_{m+2} + \dots + (c_n - z_n)x_n$$

where

$$z_0 = \bar{a}_{10}c_1 + \bar{a}_{20}c_2 + \dots + \bar{a}_{m0}c_m \quad z_j = \bar{a}_{1j}c_1 + \bar{a}_{2j}c_2 + \dots + \bar{a}_{mj}c_m \quad \text{for } m+1 \leq j \leq n$$

which gives the fundamental relation required to determine the pivot column. We note that  $z_0$  gives the value of the objective function for the current basic solution. The important point is that this equation gives the values of the objective function  $z$  for any solution of  $Ax = b$  in terms of the variables  $x_{m+1}, \dots, x_n$ . From it we can determine if there is any advantage in changing the basic solution by introducing one of the nonbasic variables. For example, if  $(c_j - z_j)$  is positive for some  $j$ ,  $m+1 \leq j \leq n$ , then increasing  $x_j$  from zero to some positive value would increase the value of the objective function, and therefore would yield a better solution.

Indeed, we may deduce :

$$\sum_{i=1}^m c_i x_i = c_B^T \bar{a}_0 - \sum_{j=m+1}^n z_j x_j,$$

where  $z_j = c_B^T \bar{a}_j$ . Thus, adding  $\sum_{j=m+1}^n c_j x_j$  to both sides we obtain:  $c^T x = z_0 + \sum_{j=m+1}^n (c_j - z_j)x_j$ .

**Theorem 3.** *Given a nondegenerate basic feasible solution with corresponding objective value  $z_0$ , suppose that for some  $j$  there holds  $(c_j - z_j) > 0$ . Then*

there is a feasible solution with objective value  $z > z_0$ . If the column  $a_j$  can be substituted for some vector in the original basis to yield a new basic feasible solution, this new solution will have  $z > z_0$ . If  $a_j$  cannot be substituted to yield a basic feasible solution, then the solution set  $K$  is unbounded and the objective function can be made arbitrarily large (toward infinity).

Optimality Condition Theorem.

**Theorem 4.** *If for some basic feasible solution  $c_j - z_j \leq 0$  for all  $j$ , then that solution is optimal to system (3.1).*

Let denote  $r_j = c_j - z_j$ . These are called relative cost coefficients.

The **simplex algorithm** can be summarized by the following steps:

- Step 0. Form a tableau corresponding to a basic feasible solution. The relative cost coefficients can be found by row reduction.
- Step 1. If each  $r_j \leq 0$ , stop; the current basic feasible solution is optimal.
- Step 2. Select  $q$  such that  $r_q > 0$  to determine which nonbasic variable is to become basic.
- Step 3. Calculate the ratios  $\bar{a}_{i0}/\bar{a}_{iq}$  for  $\bar{a}_{i,q} > 0, i = 1, 2, \dots, m$ . If no  $\bar{a}_{i,q} > 0$ , stop; the problem is unbounded. Otherwise, select  $p$  as the index  $i$  corresponding to the minimum ratio.
- Step 4. Pivot on the  $pq^{th}$  element, updating all rows including the last. Return to step 1.

Example 1.

$$\max 3x_1 + x_2 + 3x_3 \quad (3.5a)$$

*subject to*

$$2x_1 + x_2 + x_3 \leq 2 \quad (3.5b)$$

$$x_1 + 2x_2 + 3x_3 \leq 5 \quad (3.5c)$$

$$2x_1 + 2x_2 + x_3 \leq 6 \quad (3.5d)$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0. \quad (3.5e)$$

After adding slack variables  $x_4, x_5, x_6$  we obtain the first canonical table representation of the standard formulation as follows:

|       | $x_1$    | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ | $b$ |
|-------|----------|-------|-------|-------|-------|-------|-----|
| $x_4$ | <b>2</b> | 1     | 1     | 1     | 0     | 0     | 2   |
| $x_5$ | 1        | 2     | 3     | 0     | 1     | 0     | 5   |
| $x_6$ | 2        | 2     | 1     | 0     | 0     | 1     | 6   |
|       | 3        | 1     | 3     | 0     | 0     | 0     | 0   |

At this stage we choose to enter  $x_1$  and remove  $x_4$  in the current basis.

|       | $x_1$ | $x_2$ | $x_3$      | $x_4$ | $x_5$ | $x_6$ | $b$ |
|-------|-------|-------|------------|-------|-------|-------|-----|
| $x_1$ | 1     | 1/2   | 1/2        | 1/2   | 0     | 0     | 1   |
| $x_5$ | 0     | 3/2   | <b>5/2</b> | -1/2  | 1     | 0     | 4   |
| $x_6$ | 0     | 1     | 0          | -1    | 0     | 1     | 4   |
|       | 0     | -1/2  | 3/2        | -3/2  | 0     | 0     | -3  |

Then, we enter  $x_3$  and remove  $x_5$  in the current basis.

|       | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ | $b$   |
|-------|-------|-------|-------|-------|-------|-------|-------|
| $x_1$ | 1     | 1/5   | 0     | 3/5   | -1/5  | 0     | 1/5   |
| $x_3$ | 0     | 3/5   | 1     | -1/5  | 2/5   | 0     | 8/5   |
| $x_6$ | 0     | 1     | 0     | -1    | 0     | 1     | 4     |
|       | 0     | -7/5  | 0     | -6/5  | -3/5  | 0     | -27/5 |

The optimality condition is achieved in the above table and we can stop pivoting. Then, we can deduce that the solution is  $(1/5, 0, 8/5, 0, 0, 4)$  which gives  $27/5$  in the objective function.

### Exercises

$$\max 3x_1 + 3x_2 \quad (3.6a)$$

$$x_1 + 4x_2 \leq 12 \quad (3.6b)$$

$$2x_1 + x_2 \leq 10 \quad (3.6c)$$

$$x_1 \geq 0, x_2 \geq 0. \quad (3.6d)$$

Solution :  $x_1 = 4; x_2 = 2$

$$\max 2x_2 - x_1 \quad (3.7a)$$

$$-4x_1 + 6x_2 \leq 9 \quad (3.7b)$$

$$x_1 + x_2 \leq 4 \quad (3.7c)$$

$$x_1 \geq 0, x_2 \geq 0. \quad (3.7d)$$

Solution :  $x_1 = 15/10; x_2 = 25/10$

$$\max 3x_1 + 4x_2 \quad (3.8a)$$

$$x_1 + 2x_2 \leq 18 \quad (3.8b)$$

$$x_1 + x_2 \leq 10 \quad (3.8c)$$

$$3x_1 + x_2 \leq 20 \quad (3.8d)$$

$$x_1 \geq 0, x_2 \geq 0. \quad (3.8e)$$

Solution :  $x_1 = 2; x_2 = 8$

$$\max 4x_1 + 2x_2 + 3x_3 \quad (3.9a)$$

$$x_1 + x_2 + 2x_3 \leq 8 \quad (3.9b)$$

$$2x_1 + x_2 + x_3 \leq 8 \quad (3.9c)$$

$$x_1 + x_2 + x_3 \leq 5 \quad (3.9d)$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0. \quad (3.9e)$$

Solution :  $x_1 = 3; x_2 = 0, x_3 = 2$ .

### 3.1.2 Finding an initial feasible basis

A basic feasible solution is sometimes immediately available for linear programs. For example, in problems with constraints of the form  $Ax \leq b; x \geq 0$  with  $b \geq 0$ , a basic feasible solution to the corresponding standard form of the problem is provided by the slack variables.

By elementary straightforward operations the constraints of a linear programming problem can always be expressed in the form  $Ax = b; x \geq 0$  with  $b \geq 0$ . In order to find a solution to this problem consider the (artificial) minimization problem:

$$\min \sum_{i=1}^m y_i \quad (3.10a)$$

subject to

$$Ax + y = b \quad (3.10b)$$

$$x \geq 0, y \geq 0. \quad (3.10c)$$

where  $y = (y_1, y_2, \dots, y_m)$  is a vector of artificial variables. If there is a feasible solution to  $Ax = b; x \geq 0$  with  $b \geq 0$ , then it is clear that system (3.10) has a minimum value of zero with  $y = 0$ . Clearly, if the minimum value of (3.10) is zero, then the final basic solution will have all  $y_i = 0$ .

Using artificial variables, we attack a general linear programming problem by use of the two-phase method. This method consists simply of a phase I in which artificial variables are introduced as above and a basic feasible solution is found (or it is determined that no feasible solutions exist); and a phase II in which, using the basic feasible solution resulting from phase I, the original objective function is minimized. During phase II the artificial variables and the objective function of phase I are omitted. Of course, in phase I artificial variables need be introduced only in those equations that do not contain slack variables.

An example. Find a basic feasible solution to:

$$\min 4x_1 + x_2 + x_3 \quad (3.11a)$$

subject to

$$2x_1 + x_2 + 2x_3 = 4 \quad (3.11b)$$

$$3x_1 + 3x_2 + x_3 = 3 \quad (3.11c)$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0. \quad (3.11d)$$

Let start with Phase I. We introduce artificial variables  $x_4 \geq 0, x_5 \geq 0$  and an objective function  $\min (x_4 + x_5) = \max (-x_4 - x_5)$ .

The initial tableau is

|       | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $b$ |
|-------|-------|-------|-------|-------|-------|-----|
| $x_4$ | 2     | 1     | 2     | 1     | 0     | 4   |
| $x_5$ | 3     | 3     | 1     | 0     | 1     | 3   |
| $c^t$ | 0     | 0     | 0     | -1    | -1    | 0   |

A basic feasible solution to the expanded system is given by the artificial variables. To initiate the simplex procedure we must update the last row so

that it has zero components under the basic variables. This yields the first tableau as follows:

|       | $x_1$    | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $b$ |
|-------|----------|-------|-------|-------|-------|-----|
| $x_4$ | 2        | 1     | 2     | 1     | 0     | 4   |
| $x_5$ | <b>3</b> | 3     | 1     | 0     | 1     | 3   |
|       | 5        | 4     | 3     | 0     | 0     | 7   |

Pivoting in the column having the most positive bottom row component as indicated, we obtain:

|       | $x_1$ | $x_2$ | $x_3$      | $x_4$ | $x_5$ | $b$ |
|-------|-------|-------|------------|-------|-------|-----|
| $x_4$ | 0     | -1    | <b>4/3</b> | 1     | -2/3  | 2   |
| $x_1$ | 1     | 1     | 1/3        | 0     | 1/3   | 1   |
|       | 0     | -1    | 4/3        | 0     | -5/3  | 2   |

and last tableau:

|       | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $b$ |
|-------|-------|-------|-------|-------|-------|-----|
| $x_3$ | 0     | -3/4  | 1     | 3/4   | -1/2  | 3/2 |
| $x_1$ | 1     | 5/4   | 0     | -1/4  | 1/2   | 1/2 |
|       | 0     | 0     | 0     | -1    | -1    | 0   |

Both of the artificial variables have been driven out of the basis, thus reducing the value of the objective function to zero and leading to the basic feasible solution to the original problem  $x_1 = 1/2, x_2 = 0, x_3 = 3/2$ .

After this Phase I, we need to run phase II in which, we will use the basic feasible solution resulting from phase I. During phase II the artificial variables and the objective function of phase I are omitted. Of course, in phase I artificial variables need be introduced only in those equations that do not contain slack variables, but first, we convert to *max* the objective function in (3.11). The initial tableau is:

|       | $x_1$ | $x_2$ | $x_3$ | $b$ |
|-------|-------|-------|-------|-----|
| $x_3$ | 0     | -3/4  | 1     | 3/2 |
| $x_1$ | 1     | 5/4   | 0     | 1/2 |
| $c^T$ | -4    | -1    | -1    | 0   |

Transforming the last row so that zeros appear in the basic (variable) columns, we have:

|       | $x_1$ | $x_2$      | $x_3$ | $b$ |
|-------|-------|------------|-------|-----|
| $x_3$ | 0     | -3/4       | 1     | 3/2 |
| $x_1$ | 1     | <b>5/4</b> | 0     | 1/2 |
|       | 0     | 13/4       | 0     | 7/2 |

Then, the second tableau is as follows:

|       | $x_1$ | $x_2$ | $x_3$ | $b$  |
|-------|-------|-------|-------|------|
| $x_3$ | 3/5   | 0     | 1     | 9/5  |
| $x_2$ | 4/5   | 1     | 0     | 2/5  |
|       | -13/5 | 0     | 0     | 11/5 |

and hence the optimal solution is  $x_1 = 0, x_2 = 2/5, x_3 = 9/5$ .

**Example 2**

Provide a feasible basic solution for the above problem (apply phase 1) and check its optimality for the initial problem.

$$\max -2x_2 + x_3 - x_4 \quad (3.12a)$$

*subject to*

$$-x_2 + x_4 = 1 \quad (3.12b)$$

$$-x_3 + x_4 = 3 \quad (3.12c)$$

$$x_i \geq 0, i = 2, 3, 4. \quad (3.12d)$$

Solution:

The reader may note that  $b$  vector has non negative values. This is important since we are looking for a basic feasible solution and the pivoting operations guarantee the positivity of vector  $b$  throughout the simplex algorithm.

|       | $x_2$ | $x_3$ | $x_4$    | $y_1$ | $y_2$ | $b$ |
|-------|-------|-------|----------|-------|-------|-----|
| $y_1$ | -1    | 0     | <b>1</b> | 1     | 0     | 1   |
| $y_2$ | 0     | -1    | 1        | 0     | 1     | 3   |
|       | 0     | 0     | 0        | -1    | -1    | 0   |
|       | -1    | -1    | 2        | 0     | 0     | 4   |

|       | $x_2$    | $x_3$ | $x_4$ | $y_1$ | $y_2$ | $b$ |
|-------|----------|-------|-------|-------|-------|-----|
| $x_4$ | -1       | 0     | 1     | 1     | 0     | 1   |
| $y_2$ | <b>1</b> | -1    | 0     | -1    | 1     | 2   |
|       | 1        | -1    | 0     | -2    | 0     | 2   |

|       | $x_2$ | $x_3$ | $x_4$ | $y_1$ | $y_2$ | $b$ |
|-------|-------|-------|-------|-------|-------|-----|
| $x_4$ | 0     | -1    | 1     | 0     | 1     | 3   |
| $x_2$ | 1     | -1    | 0     | -1    | 1     | 2   |
|       | 0     | 0     | 0     | -1    | -1    | 0   |

From above we deduce that the optimal solution is  $(x_2 = 2, x_3 = 0, x_4 = 3)$  and the objective function value is 0. This gives a basic feasible solution for the initial problem. Next, second phase starts with the above feasible basic solution:

|       | $x_2$ | $x_3$ | $x_4$ | $b$ |
|-------|-------|-------|-------|-----|
| $x_4$ | 0     | -1    | 1     | 3   |
| $x_2$ | 1     | -1    | 0     | 2   |
|       | -2    | 1     | -1    | 0   |
|       | 0     | -2    | 0     | 7   |

We remark here that the solution is optimal for the initial problem.

**Exercise**

Find a basic feasible basis for the following problem.

$$\min x_1 + 6x_2 - 7x_3 + x_4 + 5x_5 \quad (3.13a)$$

*subject to*

$$5x_1 - 4x_2 + 13x_3 - 2x_4 + x_5 = 20 \quad (3.13b)$$

$$x_1 - x_2 + 5x_3 - x_4 + x_5 = 8 \quad (3.13c)$$

$$x_i \geq 0, i = 1, 2, 3, 4, 5. \quad (3.13d)$$

### 3.1.3 The revised simplex method

Although the elementary pivot transformations associated with the simplex method are in many respects most easily discernible in the tableau format, with attention focused on the individual elements, there is much insight to be gained by studying a matrix interpretation of the procedure. The vector-matrix relationships that exist between the various rows and columns of the tableau lead, however, not only to increased understanding but also, in a rather direct way, to the revised simplex procedure which in many cases can result in considerable computational advantage. A preliminary observation in the development is that the tableau at any point in the simplex procedure can be determined solely by a knowledge of which variables are basic. As before we denote by  $B$  the submatrix of the original  $A$  matrix consisting of the  $m$  columns of  $A$  corresponding to the basic variables. These columns are linearly independent and hence the columns of  $B$  form a basis. We refer to  $B$  as the basis matrix.

As usual, let us assume that  $B$  consists of the first  $m$  columns of  $A$ . Then by partitioning  $A$ ,  $x$ , and  $c^T$  as

$$A = [B, D], x = (x_B, x_D), \text{ and } c^T = [c_B^T, c_D^T],$$

the standard linear program becomes

$$\max c_B^T x_B + c_D^T x_D \quad (3.14a)$$

*subject to*

$$Bx_B + Dx_D = b \quad (3.14b)$$

$$x_B \geq 0, x_D \geq 0. \quad (3.14c)$$

The basic solution, which we assume is also feasible, corresponding to the basis  $B$  is  $x = (x_B, 0)$  where  $x_B = B^{-1}b$ . The basic solution results from setting  $x_D = 0$ . However, for any value of  $x_D$  the necessary value of  $x_B$  can be computed from above as

$$x_B = B^{-1}b - B^{-1}Dx_D$$

and this general expression when substituted in the cost function yields:

$$z = c_B^T(B^{-1}b - B^{-1}Dx_D) + c_D^T x_D = c_B^T B^{-1}b + (c_D^T - c_B^T B^{-1}D)x_D,$$

which expresses the cost of any solution of system (3.14) in terms of  $x_D$ . Thus  $r_D^T = c_D^T - c_B^T B^{-1}D$  is the relative cost vector (for nonbasic variables).

It is the components of this vector that are used to determine which vector to bring into the basis.

Having derived the vector expression for the relative cost it is now possible to write the simplex tableau in matrix form. The initial tableau takes the form:

$$\begin{bmatrix} A & b \\ c^T & 0 \end{bmatrix} = \begin{bmatrix} B & D & b \\ c_B^T & c_D^T & 0 \end{bmatrix}$$

which is not in general in canonical form and does not correspond to a point in the simplex procedure. If the matrix  $B$  is used as a basis, then the corresponding tableau becomes:

$$\left[ \begin{array}{c|c|c} I & B^{-1}D & B^{-1}b \\ \hline - & - & - \\ 0 & c_D^T - c_B^T B^{-1}D & -c_B^T B^{-1}b \end{array} \right]$$

which is the matrix form we desire.

#### The revised form of the simplex method.

We may note that if  $m$  is much smaller than  $n$ , that is, if the matrix  $A$  has far fewer rows than columns, pivots will occur in only a small fraction of the columns during the course of optimization. Since the other columns are not explicitly used, it appears that the work expended in calculating the elements in these columns after each pivot is, in some sense, wasted effort. The revised simplex method is a scheme for ordering the computations required of the simplex method so that unnecessary calculations are avoided. In fact, even if pivoting is eventually required in all columns, but  $m$  is small compared to  $n$ , the revised simplex method can frequently save computational effort.

The revised form of the simplex method can be formulated as follows :  
Given the problem  $\max C^T x$  subject to:  $Ax \geq b, x \geq 0$ , let  $B^{-1}$  be the inverse of a current basis, and the current solution  $x_B = \bar{a}_0 = B^{-1}b$ ,

- Step 1. Calculate the current relative cost coefficients  $r_D^T = c_D^T - c_B^T B^{-1}D$ . This can best be done by first calculating  $y^T = c_B^T B^{-1}$  and then the relative cost vector  $r_D^T = c_D^T - y^T D$ . If  $r_D \leq 0$  stop; the current solution is optimal.
- Step 2. Determine which vector  $a_q$  (with respect to  $r_q > 0$ ) is to enter the basis by selecting one (or the most) positive cost coefficient; and calculate  $\bar{a}_q = B^{-1}a_q$  which gives the vector  $a_q$  expressed in terms of the current basis.
- Step 3. If no  $\bar{a}_{iq} > 0$ , stop; the problem is unbounded. Otherwise, calculate the ratios  $\bar{a}_{i0}/\bar{a}_{iq}$  for  $\bar{a}_{iq} > 0$  and take the minimum index to determine which vector is to leave the basis.
- Step 4. Update  $B^{-1}$  and the current solution  $B^{-1}b$ . Return to Step 1.

Note that above we are dealing with a maximization problem. The only difference with a minimization problem stands in step 1 when considering the reduced

cost which should have the opposite sign.

Updating of  $B^{-1}$  is accomplished by the usual pivot operations applied where the pivot is the appropriate element in  $\bar{a}_q$ . One way to deal with this is described in the following.

Let be given a basis  $B$ ,  $B^{-1}$ , column  $a_q$  and column  $a_p$  in  $B$  to be changed. We compute  $\bar{a}^q = B^{-1}a_q$  and write down  $[\bar{a}^q, B^{-1}]$ . Next, we run the usual pivoting operations to obtain a matrix with column  $\bar{a}^q$  modified to vector  $e_p = [0, 0, \dots, 1, \dots, 0]$  with  $a$  in  $p$ -th position. The last square matrix obtained corresponds to the inverse of the new basis.

**An example.** Consider again the problem:

$$\max 3x_1 + x_2 + 3x_3 \quad (3.15a)$$

*subject to*

$$2x_1 + x_2 + x_3 \leq 2 \quad (3.15b)$$

$$x_1 + 2x_2 + 3x_3 \leq 5 \quad (3.15c)$$

$$2x_1 + 2x_2 + x_3 \leq 6 \quad (3.15d)$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0. \quad (3.15e)$$

The first canonical table representation of the above formulation is as follows:

|       | $x_1$    | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ | $b$ |
|-------|----------|-------|-------|-------|-------|-------|-----|
| $x_4$ | <b>2</b> | 1     | 1     | 1     | 0     | 0     | 2   |
| $x_5$ | 1        | 2     | 3     | 0     | 1     | 0     | 5   |
| $x_6$ | 2        | 2     | 1     | 0     | 0     | 1     | 6   |
|       | 3        | 1     | 3     | 0     | 0     | 0     | 0   |

Clearly, above we have  $B$  corresponding to columns  $x_4, x_5, x_6$  represented by an identity matrix.  $B^{-1}$  is also an identity matrix while  $D$  is represented by the first three columns. Following the revised simplex algorithm we find that  $x_1$  is an entering basic variable and  $x_4$  leaving the basis. We compute the new column  $\bar{a}^1 = a^1 B^{-1} = a^1$ . We obtain  $B^{-1}$  by applying pivoting to the following tableau:

|   |   |   |   |
|---|---|---|---|
| 2 | 1 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 2 | 0 | 0 | 1 |

which gives  $B^{-1}$  :

|      |   |   |
|------|---|---|
| 1/2  | 0 | 0 |
| -1/2 | 1 | 0 |
| -1   | 0 | 1 |

$D$  being given as :

|   |   |   |
|---|---|---|
| 1 | 1 | 1 |
| 2 | 3 | 0 |
| 2 | 1 | 0 |

we obtain  $r_D^T = c_D^T - c_B^T B^{-1} D = (-1/2, 3/2, -3/2)$ .

The solution is not optimal and the new basis is obtained by introducing  $x_3$  instead of  $x_5$ . As above, we compute first  $\bar{a}_3 = B^{-1}a_3 = (1/2, 5/2, 0)$  and  $\bar{a}_0 = B^{-1}a_0 = [1, 4, 4]$  and find the pivot  $(5/2)$ . Next we do pivoting to obtain the new  $B^{-1}$ :

$$\begin{array}{|ccc|} \hline 1/2 & 1/2 & 0 & 0 \\ \mathbf{5/2} & -1/2 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ \hline \end{array}$$

which gives  $B^{-1}$  :

$$\begin{array}{|ccc|} \hline 3/5 & -1/5 & 0 \\ -1/5 & 2/5 & 0 \\ -1 & 0 & 1 \\ \hline \end{array}$$

at the end we obtain  $r_D^T = c_D^T - c_B^T B^{-1} D = (-7/5, -6/5, -3/5)$ , which shows the optimality of the current basic solution. The optimal solution is computed as  $B^{-1}b$  which gives  $(1/5, 8/5, 4)$ .



# Chapter 4

## TD

For further exercises look at:

- Luenberger, D. G., & Ye, Y. (2015). Linear and nonlinear programming (Vol. 228). Springer.
- Vanderbei, R. J. (2015). Linear programming. Springer.

freely available on web.

## TD 1

### 1. Production

One factory produces  $P_1$  and  $P_2$ . Each of them needs a certain number of hours on machine  $A$  and  $B$ . Each machine can run over a given number of hours (which can be fractioned), but at one time only one product can be run (for instance, one machine spends 2h producing  $P_1$ , and next 2h producing  $P_2$ ). All data are given in the following table:

Table 4.1: Availability and capacity of machines  $A$  and  $B$

|                  | $A$ | $B$ |
|------------------|-----|-----|
| Demand $h/P_1$   | 1   | 3   |
| Demand $h/P_2$   | 1   | 1   |
| Availability (h) | 6   | 12  |

1. Write a LP problem.
2. Draw the geometrical space defined by the constraints.
3. Give two feasible solutions. If  $P_1$  is sold 1000 euros and  $P_2$  3000 euros, evaluate these solutions. Show that all convex combination of these two solutions is feasible and not better than each of them.

4. Put the problem under standard form. Solve the problem for all bases. Which gives the best solution? Represent it geometrically.
5. We consider now the case where  $P_1$  is sold 2000 euros and  $P_2$  1000 euros. Find the optimal solution.

## 2. To infinity and beyond!

Consider the following problem:

$$\begin{aligned} \max. \quad & x_1 + x_2 \\ \text{with} \quad & |x_1 - x_2| \leq 2 \\ & x_i \geq 0, i = 1, 2 \end{aligned}$$

1. Give the LP formulation.
2. Represent geometrically the solution space. What did you notice?
3. Put it under standard form and solve it for different basis.

## 3. Computing basis

Consider the following LP problem :

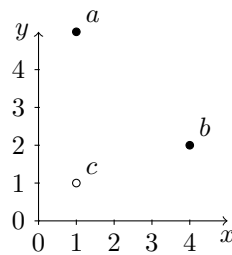
$$\left\{ \begin{array}{rcl} 2x_1 - x_2 & \leq & 4 \\ -x_1 + 2x_2 - x_3 & \leq & 8 \\ -x_2 + 2x_3 & \leq & 4 \\ x & \geq & 0 \\ cx & = & z(\max) \end{array} \right. ,$$

where  $x = (x_1, x_2, x_3)^T$ .

1. Put the problem under the standard formulation.
2. Give the basic solution corresponding to slack variables, that is  $I_1 = \{4, 5, 6\}$ .
3. Consider basis  $I_2 = \{1, 2, 3\}$ . Calculate matrix  $A^{I_2}$  and  $(A^{I_2})^{-1}$ . Give the corresponding solution ; is it feasible ? For  $b = (1, -2, 1)^T$ , the basis is always feasible?
4. Show how one can compute the inverse of  $A^{I_2}$  by using simplex pivot procedure.

## 4. Support Vector Machine

Consider points  $a = (1, 5)$ ,  $b = (4, 2)$  noted positives (+1) and point  $c = (1, 1)$  noted negative, shown below.



We need to find plan  $(w_1x + w_2y)$  which separates (with the largest  $\beta$ ) the positives points from negatives ones. We obtain:

$$\text{Min. } w_1 + w_2$$

$$w_1 + 5w_2 - \beta \geq 1$$

$$4w_1 + 2w_2 - \beta \geq 1$$

$$w_1 + w_2 - \beta \leq -1$$

$$w_1, w_2, \beta \geq 0$$

1. Put the problem under the standard formulation.
2. Solve the problem for basis  $\{4, 5, 6\}$ , what do you observe?
3. Solve the problem for basis  $\{1, 2, 3\}$ , what is the solution for  $w_1, w_2, \beta$ ?
4. Represent on the above figure the equation  $w_1x + w_2y - \beta = 0$  corresponding to this basis.

## TD 2

### 1. Production, again

One factory produces  $P_1$  and  $P_2$ . Each of them needs a certain number of hours on machine  $A$  and  $B$ . Each machine can run over a given number of hours (which can be fractionned), but at one time only one product can be run (for instance, one machine spends 2h producing  $P_1$ , and next 2h producing  $P_2$ ). All data are given in the following table:

Table 4.2: Availability and capacity of machines  $A$  et  $B$

|                  | $A$ | $B$ |
|------------------|-----|-----|
| Demande $h/P_1$  | 1   | 3   |
| Demande $h/P_2$  | 1   | 1   |
| Availability (h) | 6   | 12  |

assuming  $P_1$  is sold 2000 euros and  $P_2$  1000 euros, we are interested in maximizing the benefit.

1. Introduce the slack variables.
2. Is there any need to run the first phase?
3. Solve the problem with Simplex
4. Consider the case when  $P_1$  is sold 3000 euros and  $P_2$  1000.
5. Same question for  $P_1$  sold 4000 euros and  $P_2$  1000.

### 2. Phase 1: A (very) illustrative example

Let be given the following program:

$$\begin{aligned}
 &\min x_1 \\
 &\text{subject to:} \\
 &x_1 \leq -1, \\
 &x_i \geq 0.
 \end{aligned}$$

Obviously there is no solution, but answer to the following questions:

1. Introduce the slack variables.
2. Draw the solution space.
3. Introduce the artificial variables.
4. Solve the first phase. What do you deduce?

### 3. Unbounded

Let be given:

$$\left\{ \begin{array}{rcl} -3x_1 + 2x_2 & \leq & 4 \\ 3x_1 + 2x_2 & \leq & 16 \\ x_1 & \leq & 3 \\ x_i & \geq & 0 \\ 6x_1 + 4x_2 & = & z(\max) \end{array} \right. .$$

Show graphically that the above problem admits an infinity of solutions.

### 4. The sheriff problem

A sheriff is responsible for defining the regime for prisoners, and is given 10 euros per prisoner to do this. Nevertheless, the sheriff must follow the food regulations imposed by the government, so prisoners must, per meal:

- receive at least 30 units of nutrient A,
- receive at least 60 units of nutrient B,
- have at most 1200 calories.

The sheriff can compose his meals according to three basic foods,  $x, y, z$ , including the prices the amounts of nutrients and calories per serving are summarized in the following Table 4.3.

Table 4.3: Nutritional content and prices per nutrient

|              | $A$ | $B$ | Calories | Price |
|--------------|-----|-----|----------|-------|
| nutrient $x$ | 15  | 20  | 300      | 3     |
| nutrient $y$ | 20  | 20  | 800      | 2     |
| nutrient $z$ | 5   | 10  | 400      | 1     |

Not very concerned about the well-being of his prisoners, the Sheriff wishes to earn as much money as possible per meal, knowing that he will keep all the money unspent. We ask you:

1. Express the problem as an LP;
2. Introduce slack and artificial variables if necessary;
3. Find if the sheriff can find a solution (stage 1), and find how much he can win/lose per meal (stage 2);
4. What would happen if the regulations changed and imposed 60 units of nutrient A with a maximum of 1000 calories per meal.



# Bibliography

- [1] David G. Luenberger and Yinyu Ye. *Linear and nonlinear Programming*. Stanford University.
- [2] Robert Vanderbei. *Linear Programming - foundations and extentions*. Princeton University.
- [3] D. Bertsimas and J. Tsitsiklis. *Introduction to linear optimization*. MIT press.